

Da Vinchie Lisa

Final project for the exam of Unsupervised
Learning



MANIFOLD SCULPTING



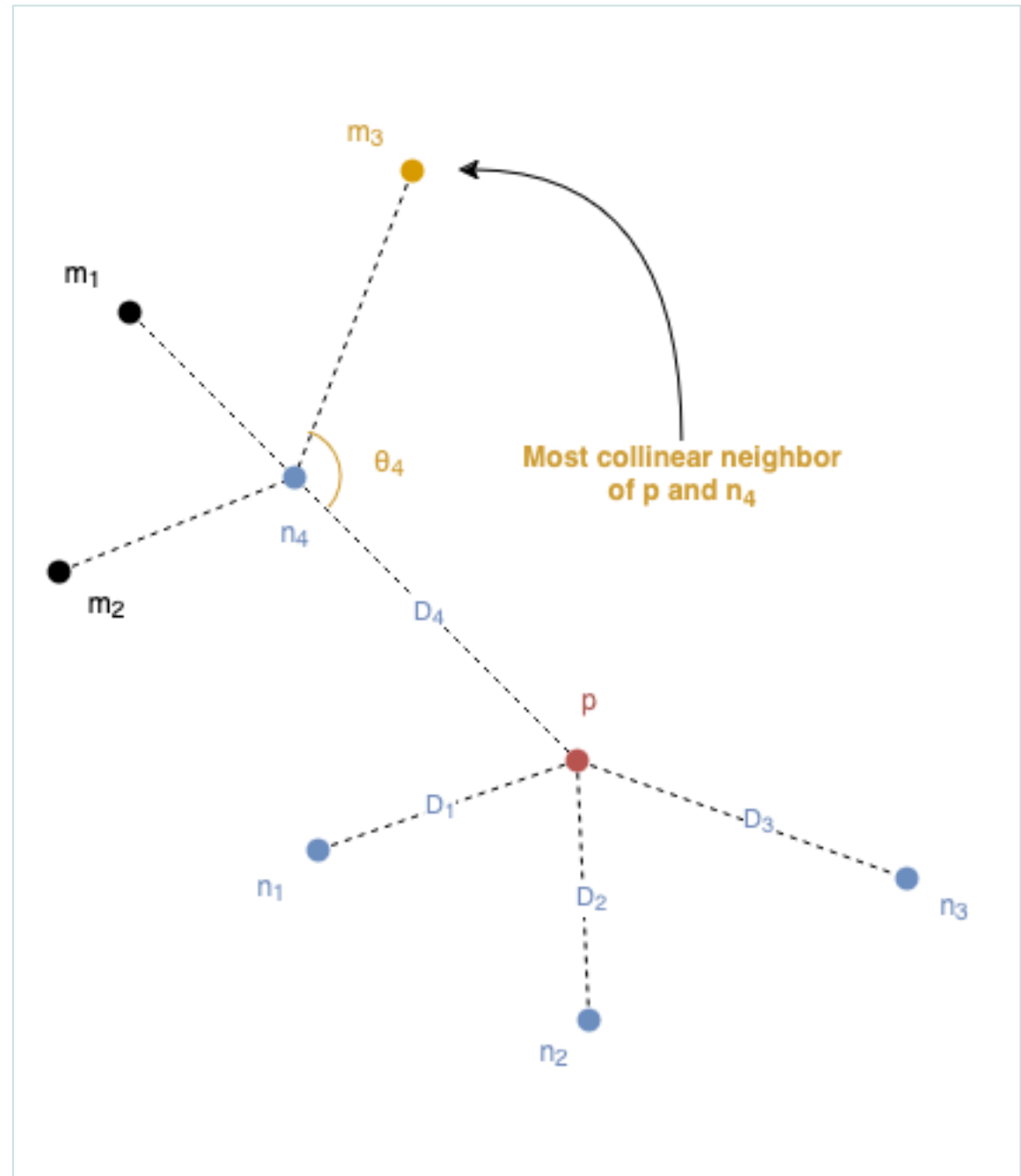
INTRO

- Non linear dimensionality reduction technique
- Preserves distances and angles between neighbors
- Can be used both in an unsupervised and semi-supervised manner

RELATIONSHIP

The relationships that we want to preserve:

- Distances D_i between each point p and its k nearest neighbors
- Angles θ_i between p , its i -th neighbor and its most collinear neighbor



LOSS

- Penalizes changes in distances and angles with most collinear neighbors
- Gives more importance to points that have already been adjusted in this step

$$L(p_i) = \sum_{j=1}^k \omega_i \left(\left(\frac{d_{ij}^0 - d_{ij}}{2 d_{avg}} \right)^2 + \left(\frac{\theta_{ij}^0 - \theta_{ij}}{\pi} \right)^2 \right)$$

ALGORITHM

1. Compute relationship between neighbors
2. Optionally perform PCA
3. Until the stopping criterion is met
 1. Scale up dimensions to preserve by σ
 2. Scale down dimensions to scale by σ
 3. Adjust preserved dimensions by shifting the points
4. Drop dimensions to scale

STOPPING CRITERION

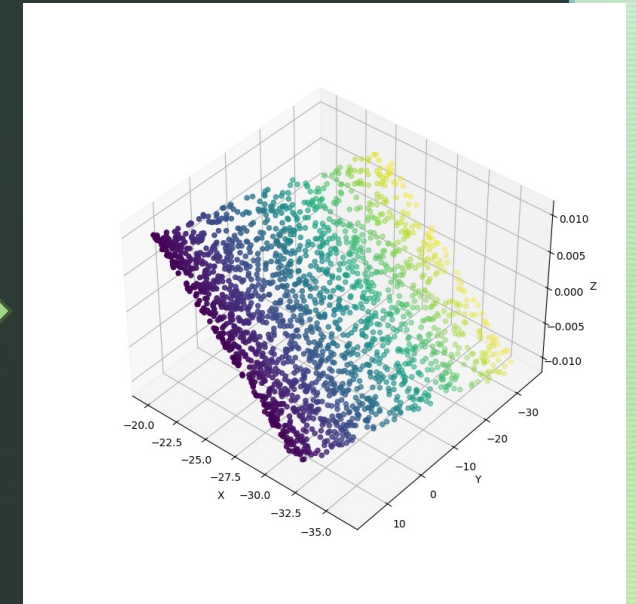
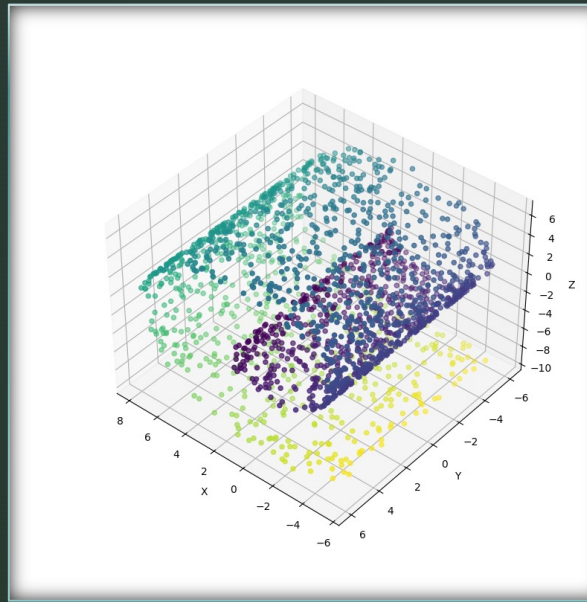
Stop when the changes between the current and previous result are small

Stop when there are no improvements in the error for some epochs

HYPERPARAMETERS

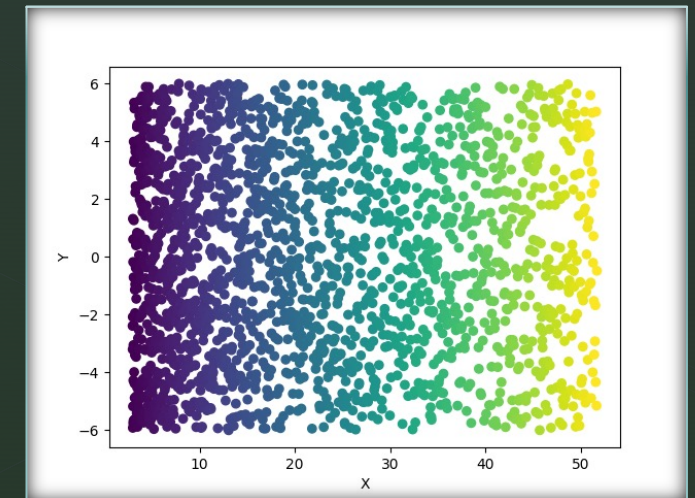
- Numbers of neighbors
 - Must be small enough to consider the surface flat
- Maximum number of iterations
 - Hard to decide a priori
- Maximum number of iterations with no error improvement
- Scale factor
 - use $\sigma = 0.99$ as indicated by the authors

VALIDATION



Made using the Swiss Roll dataset

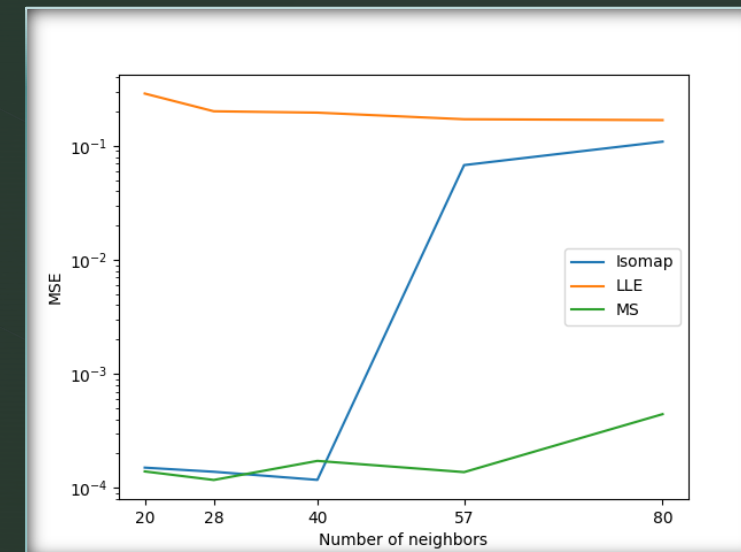
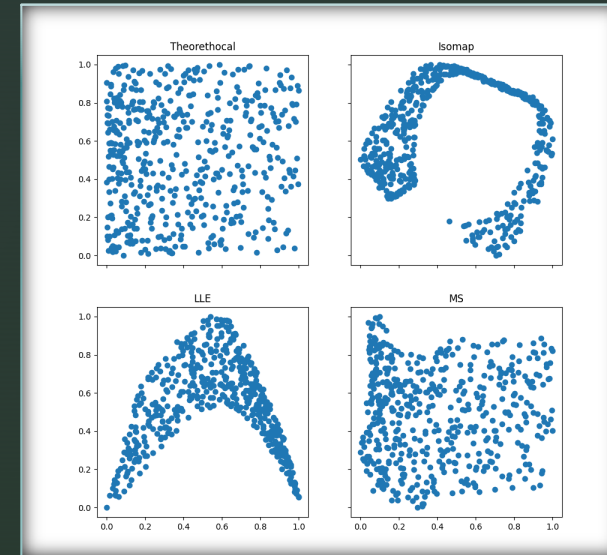
- With 2000 points and a varying number of neighbors
- With a varying density and 14 neighbors
- Use theoretical result as reference



COMPARISON WITH OTHER METHODS

Pros:

- Accurate even with a small density of points (250 in the picture)
- Number of neighbors is not that influential on the result



COMPARISON WITH OTHER METHODS

Cons:

- Slow
- Many hyperparameters to choose
- Must be rotated to align with the XY plane

